Computational Science Project

Unsupervised and supervised analysis of protein sequences

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Task 1 : Pre-processing and One-hot encoding

- **Protein MSA**, each of length **96**
- 20 Amino acids + 1 sequence alignment (‘-’) = **21 characters**
- One-hot encoding : 1 aa or ‘-’ ⇔ 1 dimension : **all aa are equal**

‘-TSENPLLALREKISALDEKLLALLAERRELAVEVGKAKLLSHRPVRDIDRERDLLERLITLGK-AHHLDAHYITRLFQLIIEDSVLTQQALLQQH’

2016-dim vector space

Task 1 : Pre-processing and One-hot encoding

- Natural data : **1130** sequences, Artificial data : **1003** sequences

  ⇒ **High dimensional** feature space, while very **small dataset**

- Rescaling data to give the same importance to each feature

  ⇒ **Centered** and **normalized** data : ready to be analysed

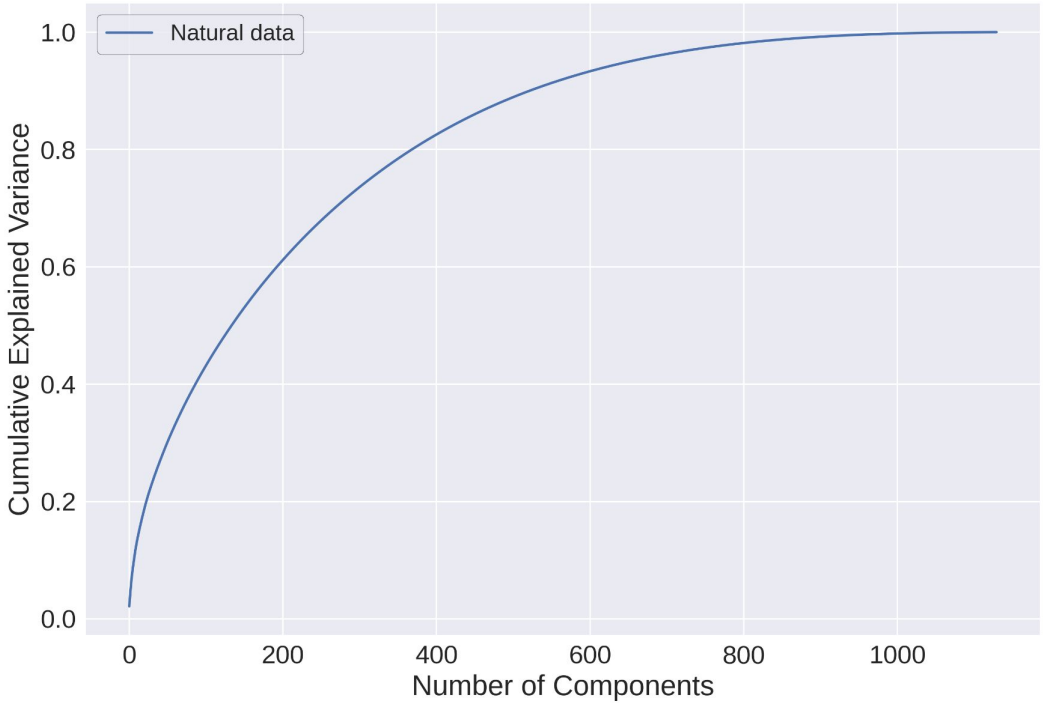
Task 2 : Principal Component Analysis (PCA)

- Question : How to **reduce the dimensionality** of the data ?

  ⇒ **Principal component analysis** : Signal is not distributed equally among all dimensions : Diagonalize the Covariance Matrix !

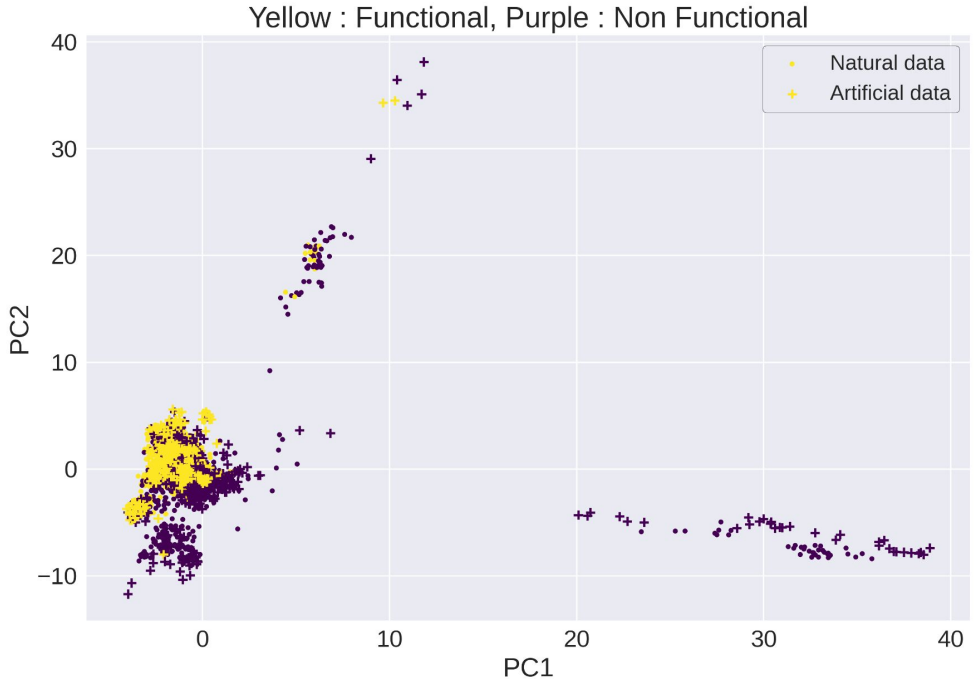
$$C_{ij} = \frac{1}{M-1} \sum_{k=1}^{M} X_i^k X_j^k$$

Cumulative explained ratio : Normalized sum of the principal components



- First two dimensions
  ⇒ **3.8 %** of the signal
- 90 % of the signal
  ⇒ **522** components needed

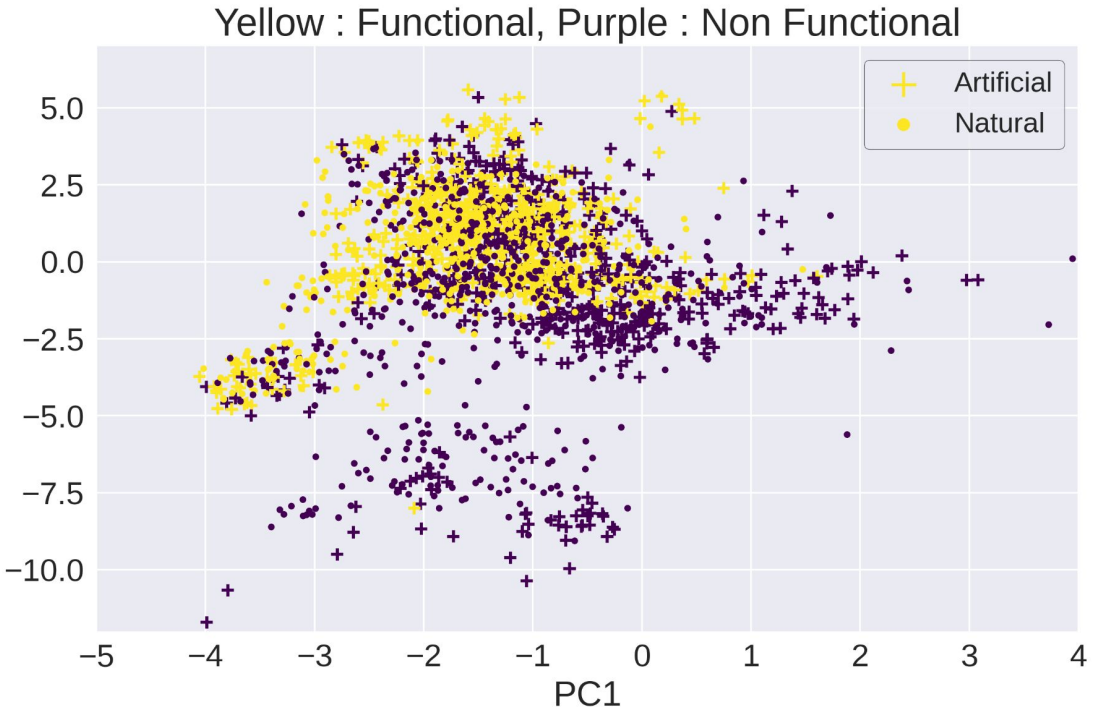
PCA space : Easy visualisation of the data



- F and NF sequences are **not well separated**, except, lower-right corner

- Natural and Artificial data **overlap**

PCA space : Easy visualisation of the data



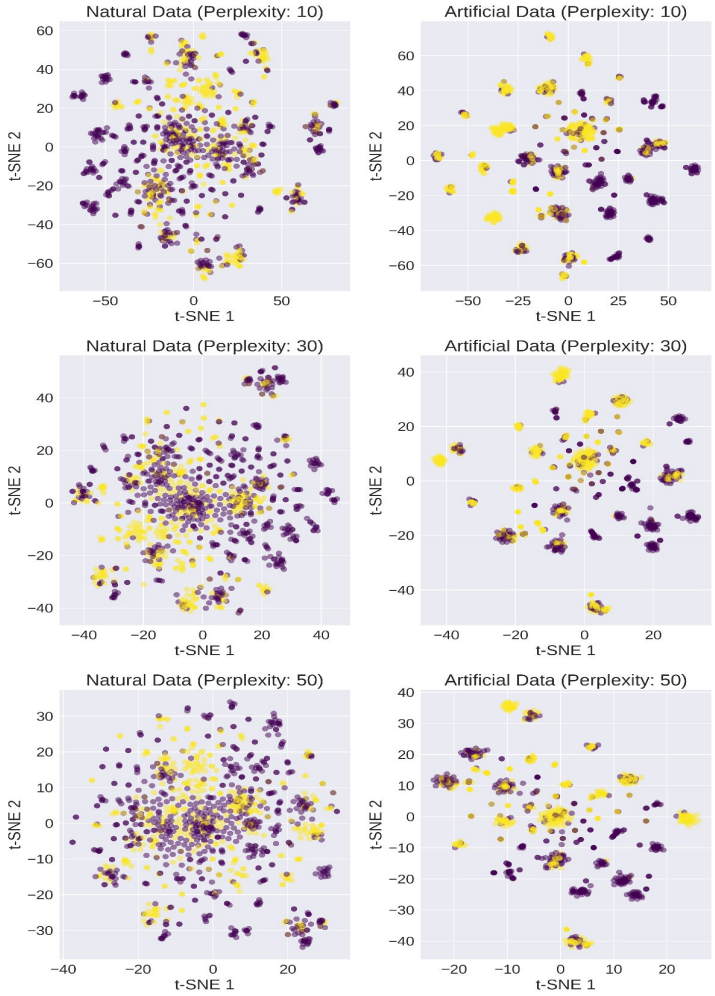
- Very dense and **mixed region** in PCA space
- PCA causes **crowding** ⇒ Not interpretable

t-SNE : Non linear mapping, preserves distances better

- Tries to **avoid crowding** : Far away points get further away and close points get closer

Short-tail neighbours distribution in feature space

Fat-tail neighbours distribution in t-SNE space



t-SNE : Non linear mapping, preserves distances better

- High computational cost : First apply PCA to help (with 800 first PCs)
- Not an explicit mapping, not reusable, non convex optimisation
- F and NF are more separated in the artificial dataset

What have we learned ?

- Even t-SNE is not able to separate F and NF sequences

  ⇒ F and NF sequences look **really similar**

- Artificial data is overlapping Natural data almost everywhere

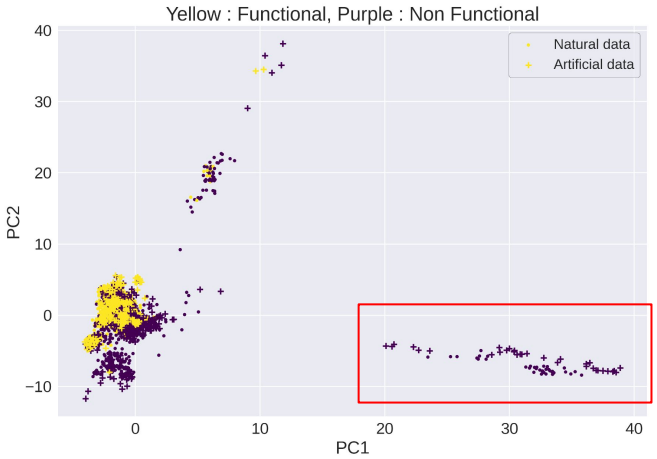
  ⇒ Insight of the **quality of the Artificial data**

Task 3 : Clustering

- Question : Are there **similar points** in the dataset ? And is it possible to **cluster F and NF sequences apart**, without knowing a priori their class ?

  - Lower right corner looks different from the rest

  - All the points in this line are NF sequences



Task 3 : A) k-means algorithm

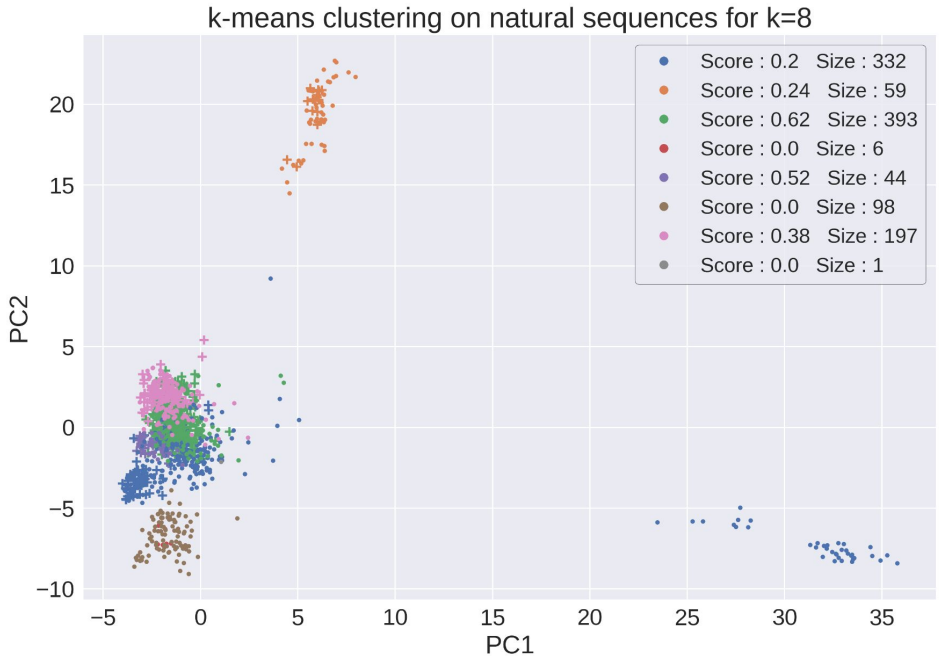
- Trying to fit **k cluster centers**, and **cluster assignation** for each datapoint, by minimising the distance between a point and its cluster center
- We define the **Score** of a cluster :

$$\text{Score} = \frac{\text{Number of F sequences in the cluster}}{\text{Total number of sequences in the cluster}}$$

⇒ 0 or 1 score means separated cluster

Task 3 : A) k-means algorithm

- **k** : Hyperparameter to fit
- Assumes **spherical clusters**
- Allows a **single point** to be a cluster



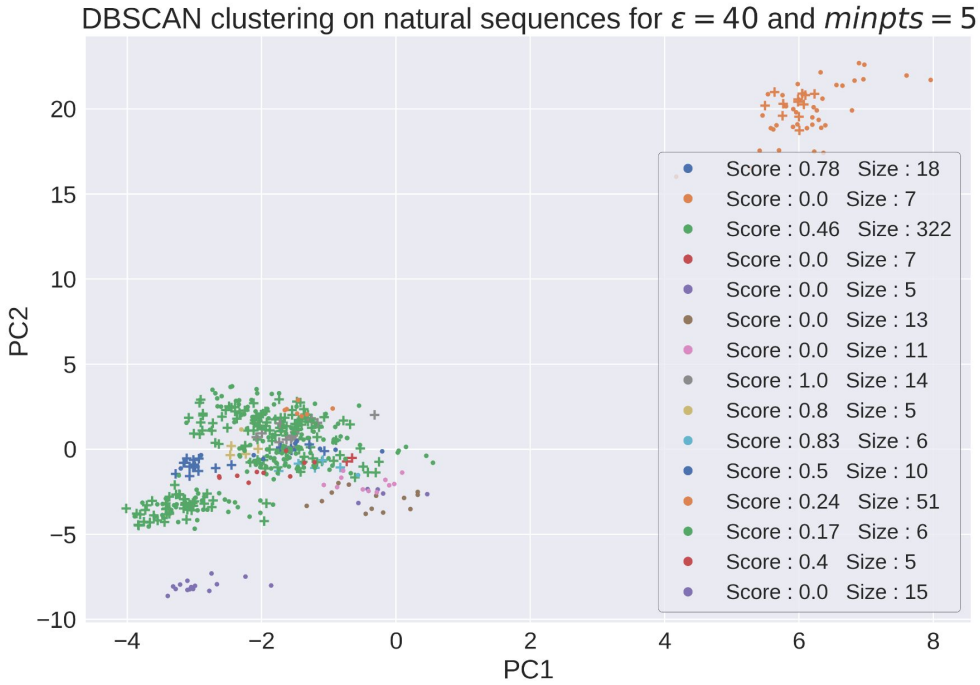
Task 3 : B) Density Based Scan

- Trying to find **core points** from which you can access the other points in the cluster via a path
- Maximum step size of the path : $\mathcal{E}$
- Minimum number of points to be a cluster : $minpts$

⇒ Two hyperparameters to choose

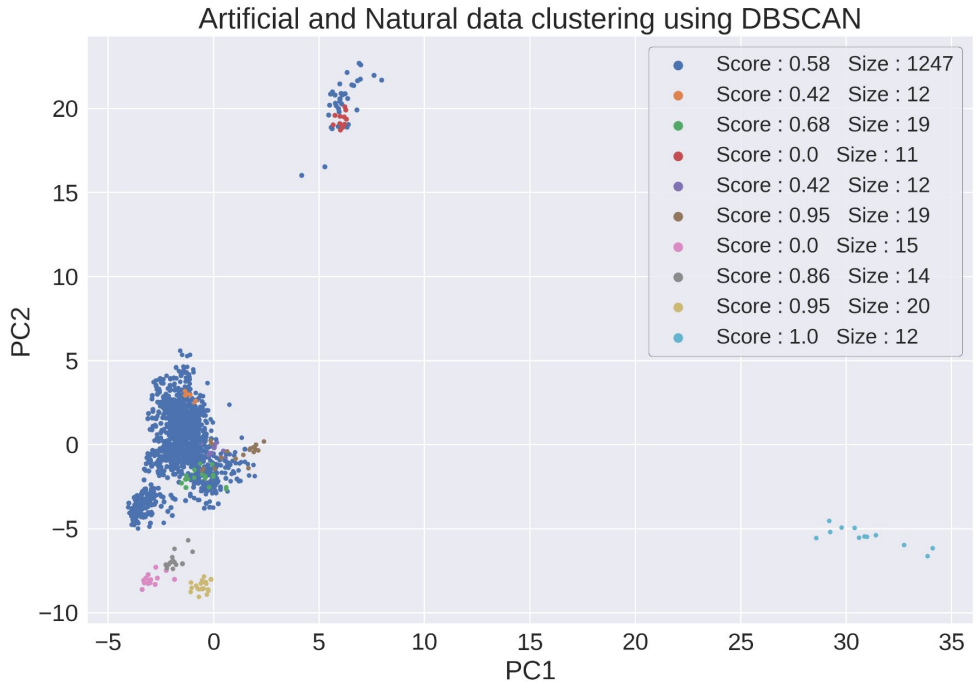
- $\mathcal{E}$ **too small** divides similar data, $\mathcal{E}$ **too big** clusters different points.

Task 3 : B) Density Based Scan



- More 0 than 1 : F are not easy to isolate
- Giant cluster of size 322 ⇒ ε should vary in different areas of feature space
- **15** clusters approx.

Task 3 : C) DBScan on Artificial and Natural data



- Able to pick up slight differences between the two datasets, except in the main blob

What have we learned ?

- Even DBSCAN is not able to cluster F and NF sequences apart

  ⇒ Data has a **complicated high dimensional shape**

- Artificial and Natural data are difficult to cluster apart

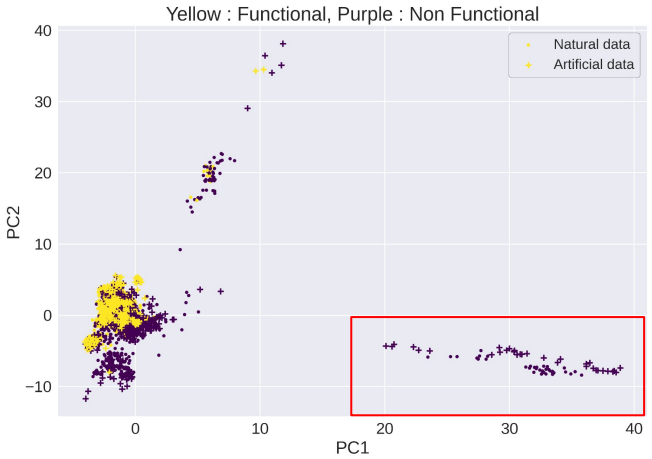
  ⇒ Another insight of the **quality of the Artificial data**

Task 4 : Classification

- Question : Given a sequence, is it possible to predict its functionality (F or NF) ?

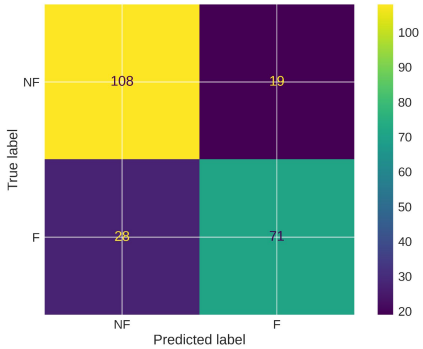
  - If we get a new point in the lower right corner, it is very likely to be a NF sequence

  - More complicated elsewhere



Task 4 : A) Logistic regression

- **Soft classifier,** trained with LASSO regularisation

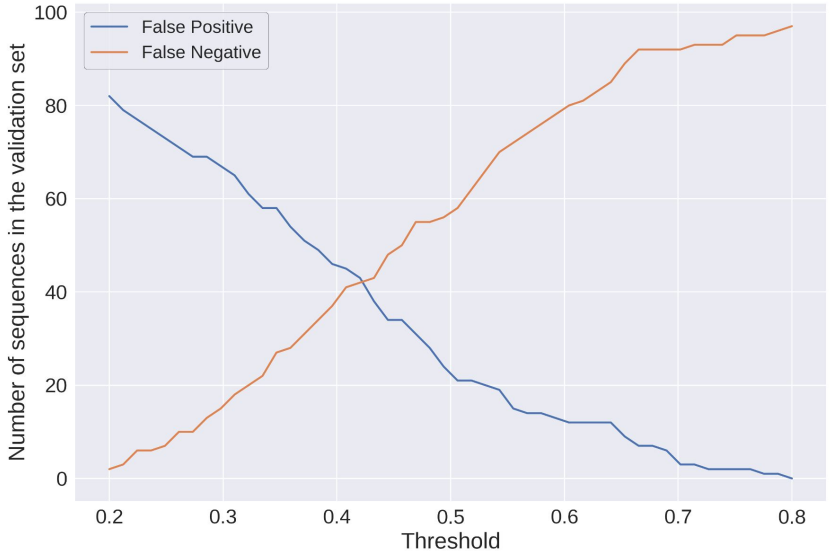


- Approx. 75% accuracy on artificial sequences. Random classifier has 50% accuracy : **Not perfect**

Task 4 : A) Logistic regression

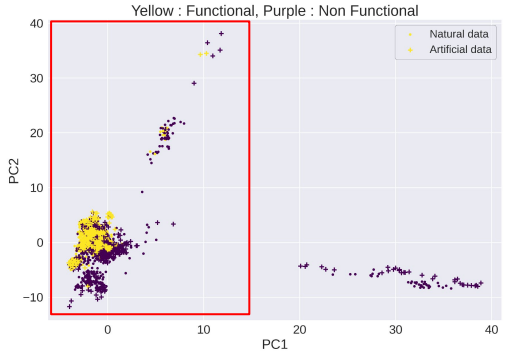
- Choosing a **threshold** for the classifier :

- Higher threshold, **more confident** of the sequences labeled as Functional
  ⇒ 0.6 - 0.7 is good



Task 4 : A) Logistic regression

- Focusing on the **main blob** of PCA space :



- Accuracy drops down to **60%** : Not an easy task

Task 4 : B) Random forest classifier

- **Ensemble method** : Avoid overfitting

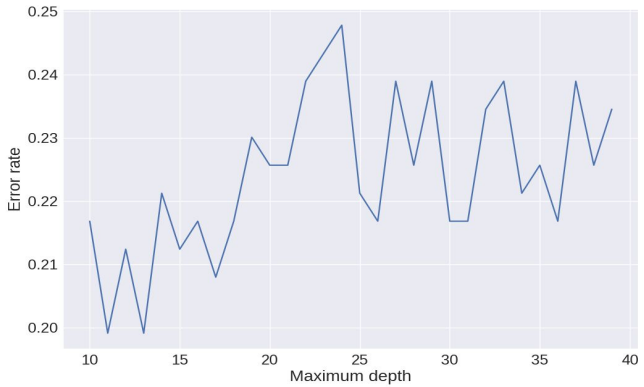
  ⇒ Good for limited data

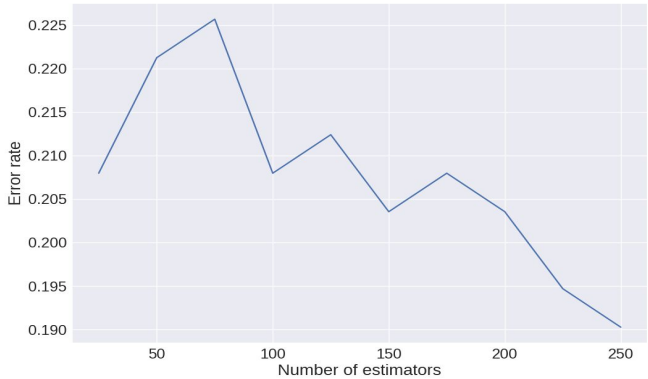
- Train a set of random decision trees, then take **majority vote**

- **Balanced** class weights in order for F and NF to be presented equally

Task 4 : B) Random forest classifier

- Validation of the **hyperparameters** : Number of trees, and maximum depth :
- We find **13** and **250** respectively





Task 4 : B) Random forest classifier

- Noisy validation curves → **k-fold** cross validation, k = 5

- Average accuracy : **80%.** Slightly better, but still some way to go

```
Average Accuracy across 5 folds: 0.796
Average Confusion Matrix:
[[113.8  27.6]
 [ 18.4  66.2]]
```

What have we learned ?

- As expected, F and NF sequences are **hard to classify**

  ⇒ Data has a complicated high dimensional shape

- However, a **simple classifier** is able to tell apart artificial and natural sequences

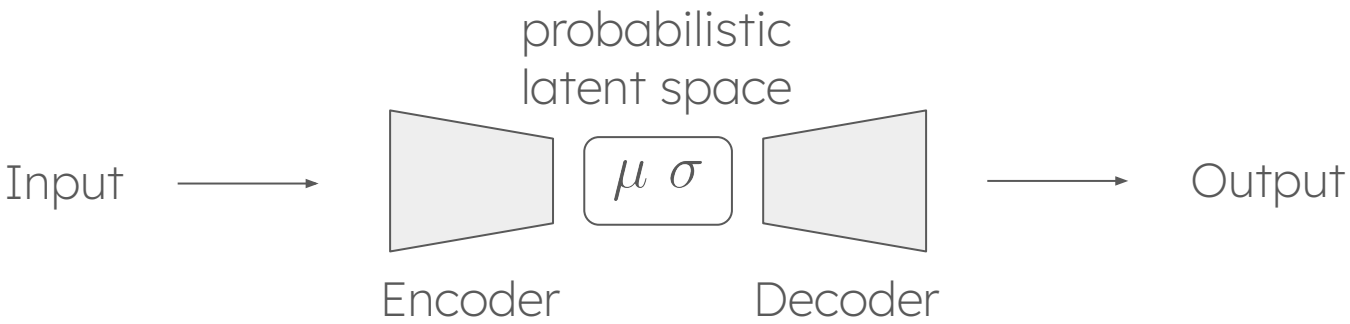
Task 5 : Generative models

- Question : Is it possible to generate new sequences that are statistically similar to the natural sequences ?

  - bmDCA is able to do it
  - But there are other ways : VAE, GAN, GMM ...

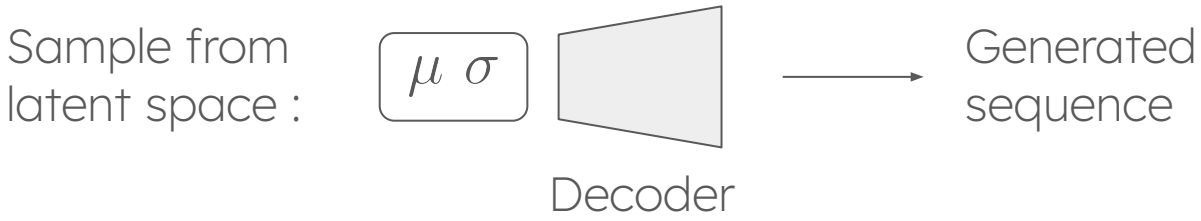
Task 5 : A) Variational Auto-encoder

- Tries to **learn the data distribution** in the latent space
- Train by minimizing the **reconstruction loss** and the dissimilarity of the two distribution in latent space (**Kullback-Leibler divergence**)

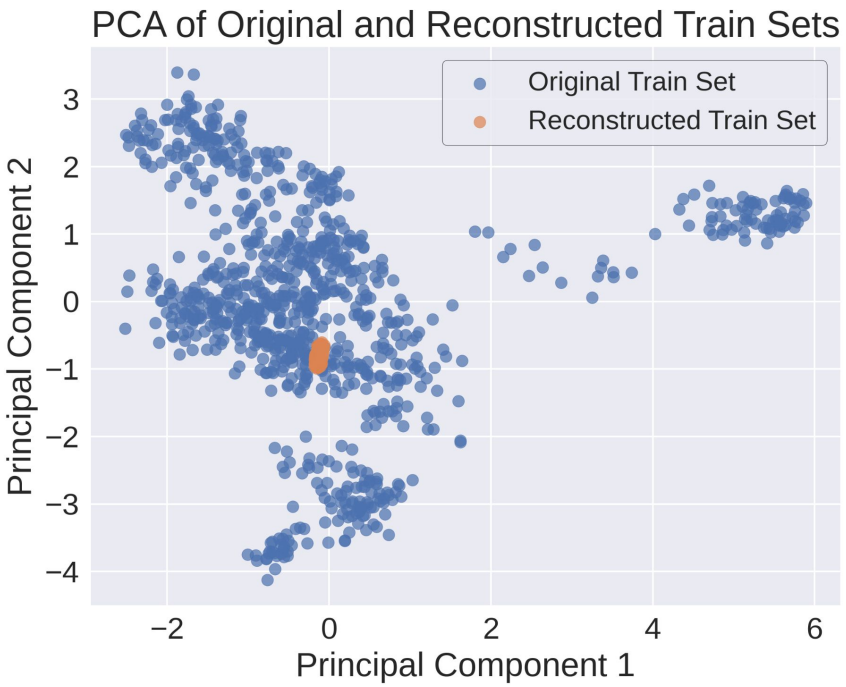


Task 5 : A) Variational Auto-encoder

- Once trained, you can generate new sequences by **sampling the prior distribution**, and then **decoding** the sample.



Task 5 : A) Variational Auto-encoder



- batch_size : 32, epochs : 10, lr : 1e-4,
- encoder : (2016→512→128), decoder : (128→512→2016)
- **Difficult to train** both prior distribution and reconstruction at the same time. Poor results even on training set

Task 5 : A) Variational Auto-encoder

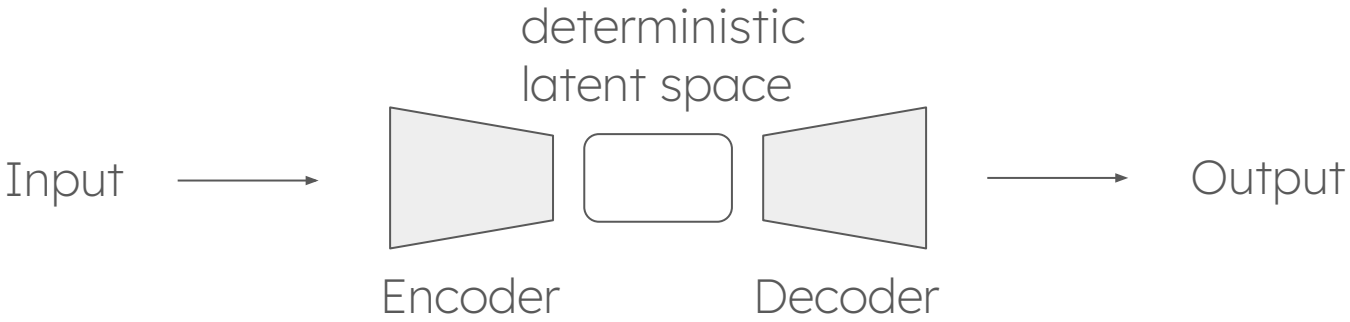
Why it did not work ?

- Minimizes the KL divergence **too quickly** compared to the reconstruction loss
- Tries to fit a **complicated** data distribution with a unimodal distribution
- Probably **bad choices** of parameters

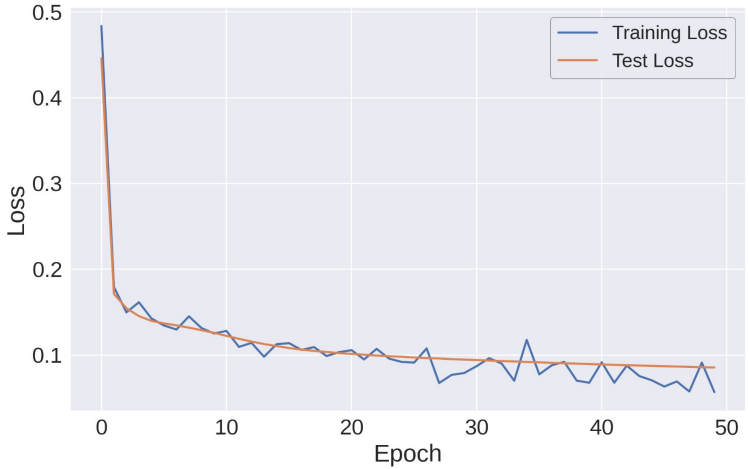
Task 5 : B) Auto-encoder and Gaussian Mixture Model

- First learn the deterministic **mapping to latent space**

- Then **train a GMM** on the encoded data to generate new sequences

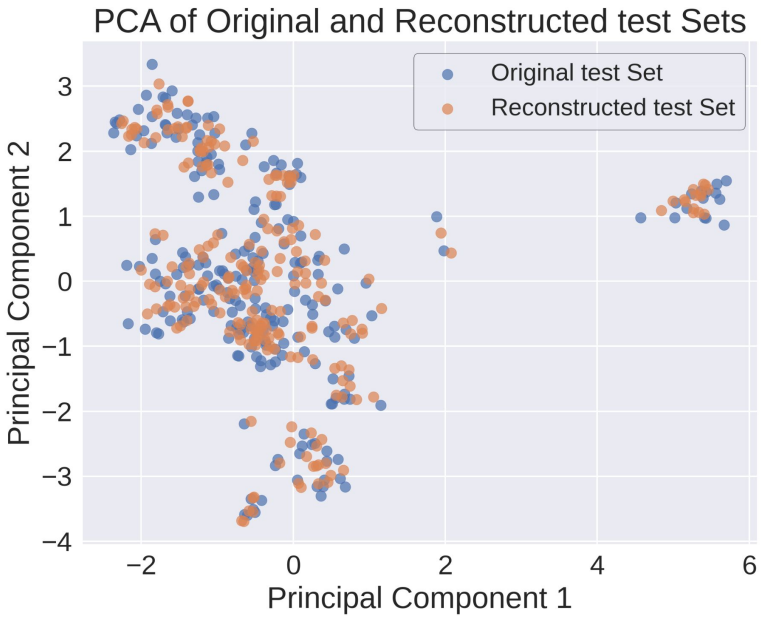


Task 5 : B) Auto-encoder and Gaussian Mixture Model



- batch_size : 100, epochs : 50, lr : 1e-3,
- encoder : (2016→128), decoder : (128→2016)
- **Generalizes well** on test sequences

Task 5 : B) Auto-encoder and Gaussian Mixture Model



- **Quick and easy** training, reproducible

- Only **9%** of the information is lost when encoding to a 128-dim. latent space, while PCA would need 500 dim.

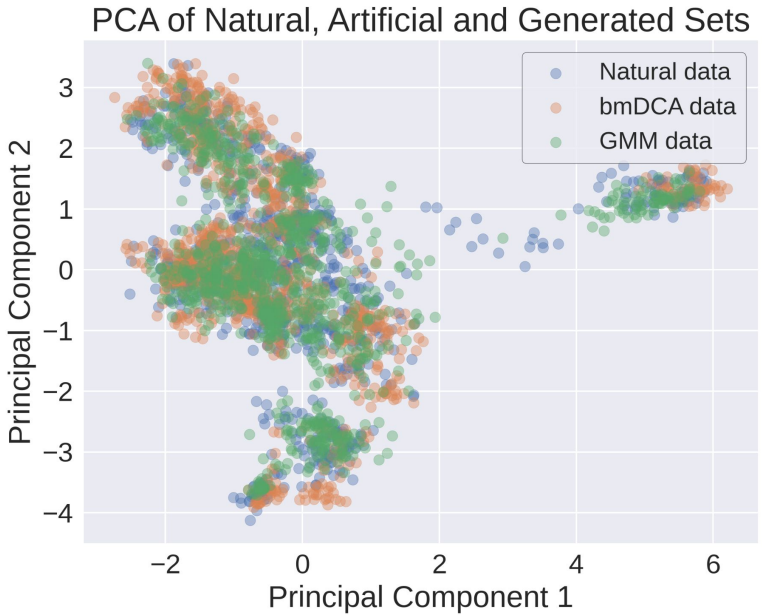
Task 5 : B) Auto-encoder and Gaussian Mixture Model

- Gaussian Mixture Model (GMM) : **clustering** algorithm, but also **generative**.

- Fits the data with a **mixture of k gaussians** of different means, variances and heights. k is hyperparameter.

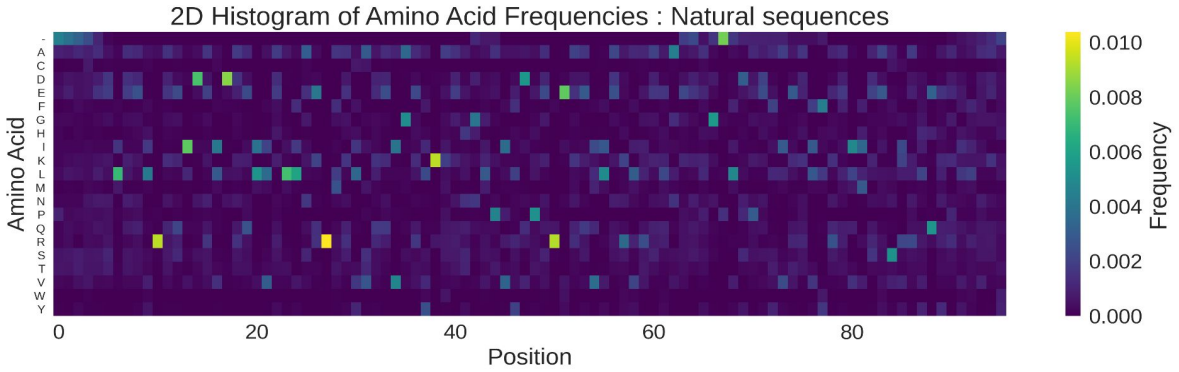
- **k=15** because it is the number of clusters given by DBSCAN earlier.

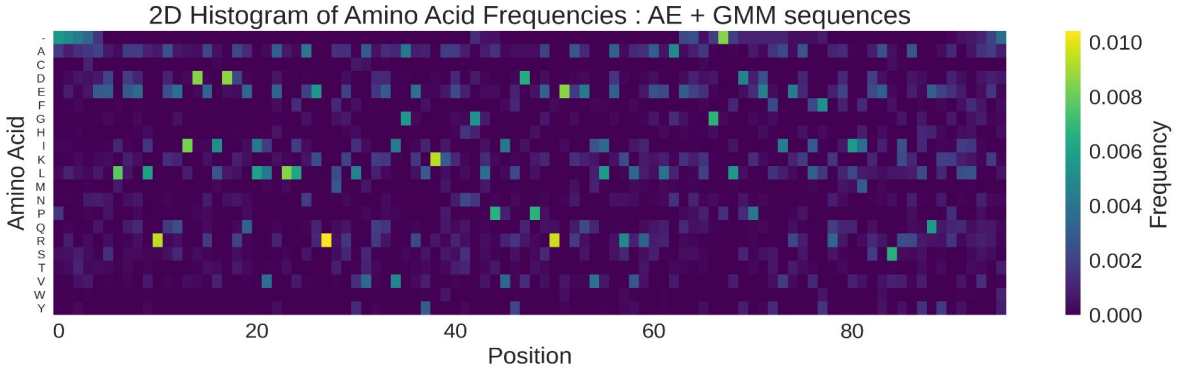
Task 5 : B) Auto-encoder and Gaussian Mixture Model



- 1st look is promising : The generated sequences occupy similar regions of PCA space

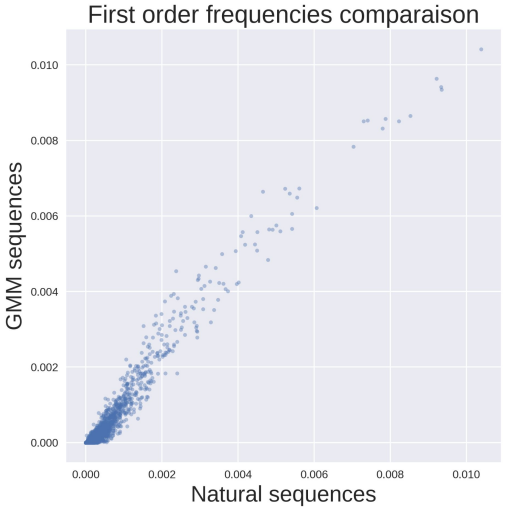
Task 5 : B) Auto-encoder and Gaussian Mixture Model

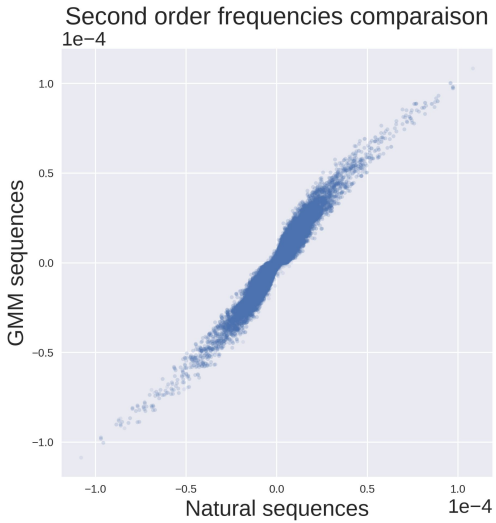




- First order statistics match

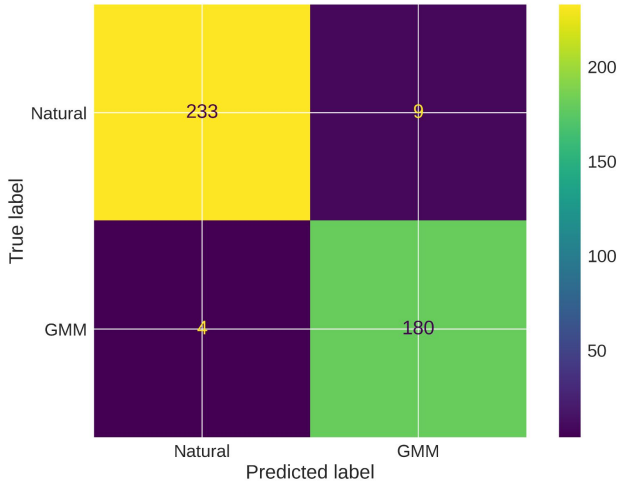
Task 5 : B) Auto-encoder and Gaussian Mixture Model





- First **and** second order statistics match : 97% and 95% corr. coeff

Task 5 : B) Auto-encoder and Gaussian Mixture Model



- However, a simple classifier is almost able to split them perfectly

What have we learned ?

- We obtained **promising results** from a very simple generative model : It presents **statistical similarities** with the original sequences, as we wanted.

- However, the generated sequences are **not realistic enough** : A simple classifier is able to tell them apart.

- **Idea** : Train a **GAN** to ensure the generated data is realistic !

Thanks for listening !